AYOUB IDEL

Senior Software Engineer | DevOps | Generative AI

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PROFILE

- **7+ years** designing, building, and operating **enterprise backend platforms** — covering **Golang development, Kubernetes, AWS cloud architecture, and Generative AI**. End-to-end ownership: from architecture to production.

At **Imagino**, a real-time **Customer Data & Engagement Platform (CDP/CEP)**, I build the backend and AI layers that let enterprises unify their data into a 360° customer view and activate it across channels instantly. On the platform side, I design and scale **Golang microservices on Kubernetes** — including a re-architecture of the real-time event pipeline that now sustains **50,000 events/second**. On the GenAI side, I ship **LLM-powered features end to end**: Claude integration, MCP servers exposing CDP data to AI agents, and a documentation RAG system.

Earlier, as **Cloud Architect & Fullstack Developer at Addict Mobile** (performance marketing / mobile user acquisition), I designed scalable, **serverless-first AWS architectures** (Lambda, Step Functions, DynamoDB) with **Infrastructure as Code in Go** (AWS CDK). I built real-time data pipelines, led the **Elasticsearch-to-OpenSearch migration**, and ran a cost-review process that meaningfully cut AWS spend.

KEY SKILLS

- **GenAI / LLM:** MCP (Model Context Protocol) servers, Claude Agent SDK, AI agents, LangChain, LangGraph, RAG (query translation — multi-query, HyDE; routing; indexing — chunk optimization, multi-representation; hybrid search, re-ranking), structured outputs (JSON Schema), prompt engineering, guardrails, Text-to-SQL, evaluation (DeepEval, Ragas)
- **LLM Ops:** AWS Bedrock, GCP Vertex AI (Gemini), Anthropic & OpenAI APIs, LiteLLM / LLM gateway, LangSmith, Langfuse, tracing/debugging, cost tracking, shadow-mode rollout, model fine-tuning
- **Infrastructure:** Kubernetes, Kubernetes Operators (Kubebuilder), Helm, Docker/Podman, ArgoCD, Terraform, AWS CDK, Ansible, GitLab CI/CD, GitHub Actions, GitOps
- **Backend:** Golang, Rust, Python (FastAPI, Pydantic), Java (Spring Boot), REST, gRPC, GraphQL
- **Frontend:** React, Angular, TypeScript, HTML/CSS
- **Cloud:** AWS (Lambda, Step Functions, DynamoDB, ECS, S3, RDS, EKS, Kinesis, SQS, Secrets Manager, Glue, OpenSearch, ...), GCP (Cloud Functions, Workflows, GKE), Azure, S3NS (GCP-based sovereign cloud)
- **Data:** PostgreSQL, DuckDB, BigQuery, DynamoDB, NoSQL, Elasticsearch/OpenSearch, ClickHouse
- **Observability:** Prometheus, Grafana, ELK, CloudWatch
- **Security:** Container & image scanning (Trivy, JFrog Xray), dependency & CVE analysis (Go: govulncheck, gosec; Java: OWASP Dependency-Check), secret & credential-leak detection in Git (Gitleaks, TruffleHog), SAST (Semgrep), SBOM
- **Blockchain:** Hyperledger Fabric, Bitcoin Core & Lightning Network (Go), go-ethereum (geth)

PROFESSIONAL EXPERIENCE

Software Engineer — Imagino

Apr 2024 – Present | Paris

*Customer Data & Engagement Platform — Imagino builds a next-gen **CDP/CEP** enabling enterprises to unify data into a **360° customer view** and activate it across channels in real time.*

- Designed and optimized **microservices & web services in Golang**, ensuring scalability and resilience.
- **I re-architected the real-time webhook & event pipeline** from a **multi-tenant to a single-tenant model** — stateless Golang microservices with per-client isolation, dedicated pipelines and independent horizontal scaling, durable event streaming with backpressure, idempotency and retry/dead-letter, on Kubernetes with autoscaling — raising sustained capacity to **50,000 events/second** with headroom to scale further.
- **I rebuilt the real-time tagging engine** (profile qualification & segmentation): simplified the evaluation chain and tech stack, moved to single-tenant isolation — easier to operate and more resilient at scale.
- **I integrated Claude and built MCP (Model Context Protocol) servers** exposing CDP data and tools to AI agents, with **structured outputs (JSON Schema)** and guardrails for controlled data access.
- **Built a RAG system over internal documentation**: query translation (**multi-query, HyDE**), query **routing**, and an optimized **indexing** pipeline (chunk-size optimization, multi-representation indexing) to improve retrieval relevance.
- Delivered new activation channels and **source-integration plugins**: added push notifications to the real-time chain and built plugins to ingest and normalize new external sources into the CDP model.
- Improved **CI/CD and Kubernetes/Helm deployments**, added per-client dev/demo provisioning, mentored developers, and conducted **technical interviews**.

Stack: Golang, Kubernetes, Helm, Azure, GCP, S3NS, BigQuery, PostgreSQL, DuckDB, ClickHouse, REST, React, Claude, MCP, LangChain, LangGraph, LangSmith, Prometheus/Grafana, stateless & highly available services

Cloud Architect & Fullstack Developer — Addict Mobile
Feb 2020 – Apr 2024 | *Paris*

Performance Marketing / Mobile User Acquisition

Cloud Architect (Mar 2022 – Apr 2024)

- **Built a real-time ad-creative generation engine** (Golang): multi-source ingestion pipelines and a **config-driven merge engine** set per account by Account Managers, feeding real-time generation orchestrated in **serverless** (Step Functions, Lambda, DynamoDB, AWS Glue) with heavier jobs offloaded to ECS — a scalable pipeline the business drives without engineering per account.
- Designed scalable **AWS backend architectures** (high availability, serverless) and built projects on **ECS Fargate** with **Infrastructure as Code in Go (AWS CDK)** for reproducible, faster deployments.
- Led the **Elasticsearch to OpenSearch migration**, improving performance while cutting costs.
- **I owned the cloud cost review process**: I cost-modelled every new project at design time, monitored daily consumption per project, and corrected drift against the estimate — cutting AWS infrastructure costs significantly.

Fullstack Developer (Feb 2020 – Mar 2022)

- Built backend services and features in **Golang**, **Python**, and **Java (Spring Boot)**.
- Built responsive front-end interfaces in **React** and **Angular**, wired to the backend REST APIs.
- Designed **REST APIs** following clean architecture principles.
- Deployed containerized services on **AWS** (ECS, S3, RDS, Lambda…).

Stack: Golang, Python, Java (Spring Boot), React, Angular, AWS, AWS CDK, Docker, serverless, OpenSearch/Elasticsearch, REST

Backend Developer — Astek
Aug 2019 – Feb 2020

- Drove the migration of internal microservices from **REST to gRPC** to improve performance, scalability, and reliability.
- Implemented the **gRPC communication layer in Golang**.

Stack: Golang, gRPC, Docker, AWS, microservices

PROJECTS

pod-go-helm-factory — Helm Chart Factory for Kubernetes
Go, Helm

- **Built a platform that programmatically generates production-grade Helm charts** for Kubernetes from **typed configuration** — application teams ship consistent, secure deployments without hand-writing YAML. **Security & reliability are baked in by default** (non-root containers, read-only filesystems, least-privilege capabilities, network isolation, pod disruption budgets, autoscaling), exposed as composable, opt-in building blocks (ingress, autoscaling, monitoring, network policy).
- **Custom deterministic rendering engine** with an automated test suite that validates every generated chart **end-to-end against real tooling**; architected for **fleet-scale multi-tenant delivery** — per-branch preview environments, layered per-environment configuration, and progressive rollouts across many customer production environments, driven by CI/CD.

Stack: Go, Kubernetes, Helm, Docker, GitHub Actions, YAML • github.com/pskshksh/pod-go-helm-factory

AI Documentation Agent — Autonomous CI/CD Agent
Python, LangChain

- Built an **autonomous Python agent** running post-merge in **GitLab CI**: reads code diffs, updates a separate docs repository, and opens a review-ready MR with the code author auto-assigned as reviewer.
- Made it **safe and measurable**: **JSON-Schema-validated structured output** with per-claim `file:line` citations, a least-privilege boundary (model only edits files; git/API actions in tested, deterministic code), and LLM Ops instrumentation (per-run cost, transcripts, shadow-mode rollout).

Stack: Python, LangChain, LangGraph, Anthropic API (Claude), GitLab CI/CD, Docker, JSON Schema, prompt engineering

Security Audit Agent — Pluggable Scanners, One Owned Backlog
Go, Claude

An agent that runs a configurable set of security tools over a repository, merges what they report into a single ranked backlog, and puts every issue in front of the person who owns the code.

- **Pluggable scanners** (JFrog Xray, secret & container scanners) behind one Go interface, run in parallel with failure isolation; heterogeneous output normalized into a single finding model and deduped by stable fingerprint.
- **Policy ranks, the model explains**: deterministic ordering (severity, license rules), while the agent clusters findings and writes the “why it matters” summary with tool + `file:line` citations via structured output.

Stack: Golang, Claude Agent SDK (headless), JFrog Xray + pluggable scanner adapters, GitLab CI/CD, Docker, JSON Schema